

Lab 6: Image Classification with CNNs

Evaluation: 5% of the total grade

In this Lab, you will build an image classification model using deep learning convolution neural networks (CNNs) using a smaller version of the cats-and-dogs dataset from the link below:

URL: 'https://storage.googleapis.com/mledu-datasets/cats_and_dogs_filtered.zip'

Perform the following Tasks:

1. Download and explore the dataset
2. Preprocess and prepare the data for modeling
3. Develop a baseline model using 1, 2, and 3 blocks as in the VGG model
4. Train and evaluate the models and report their results as discussed in the class
5. Develop a model with transfer learning using MobileNet V2 as the base model
6. Fine tune the model by unfreezing and retraining some top layers of the base model

Briefly discuss the performance of each model on an independent test set as well as a brief summary at the end comparing the performance of all the models.

An example notebook is provided to guide you through various steps in the VGG base model.

Instructions will be provided in class to guide you through transfer-learning and fine-tuning steps.

Submit your lab report along with the source code by the due date as per the course schedule.

Lab Grading Criteria

Fraction	0.8 – 1	0.4 – 0.7	0 – 0.3	Max Score
Category				
Model development workflow	Followed all steps in the modeling process	Followed most steps in the modeling process	No order in the modeling process	1
Tasks and observations	Performed all the tasks correctly, results properly presented and well explained	Performed most of the tasks correctly, results fairly presented and explained	Performed some of the tasks correctly, poorly presented results with little explanation	3
Code	Well organized, easy to follow and clean code with meaningful comments	Somewhat organized, a little hard to follow and fairly clean code with some comments	Not organized, very hard to follow and messy code with little or no comments	1
Total				5